

Experiment 1.1: Design and Performance Evaluation of a Butterworth IIR Low-Pass Filter

MATLAB Code:

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%% Experiment 1.1: Butterworth IIR Low-Pass Filter
% Design and Analysis using MATLAB

clc;
clear;
close all;

%% Given Specifications
Rp = 1;           % Passband Ripple (dB)
Rs = 20;          % Stopband Attenuation (dB)
Fp = 1000;        % Passband Frequency (Hz)
Fs_stop = 3000;   % Stopband Frequency (Hz)
Fs = 8000;        % Sampling Frequency (Hz)

%% Normalized Frequencies
Wp = Fp / (Fs/2); % Passband (normalized)
Ws = Fs_stop / (Fs/2); % Stopband (normalized)

%% Filter Order Calculation
[N, Wn] = buttord(Wp, Ws, Rp, Rs);

fprintf('Butterworth Filter Order: %d\n', N);

%% Filter Design
[b, a] = butter(N, Wn, 'low');

%% Frequency Response
[H, f] = freqz(b, a, 1024, Fs);

%% Plotting
figure('Name', 'Butterworth IIR LPF Analysis', 'NumberTitle', 'off');
tiledlayout(3,2);

nexttile;
plot(f, 20*log10(abs(H))); grid on;
title('Magnitude Response (dB)');

nexttile;
plot(f, angle(H)); grid on;
title('Phase Response');

nexttile;
[h_imp, n] = impz(b, a, 50);
stem(n, h_imp); grid on;
title('Impulse Response');

nexttile;
step_signal = ones(1,50);
step_output = filter(b, a, step_signal);
stem(0:49, step_output); grid on;
title('Step Response');

nexttile;
zplane(b, a);
title('Pole-Zero Plot');

nexttile;
plot(f, abs(H)); grid on;
title('Magnitude (Linear)');
```

Output:

